



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO LAKSHMI PETROLEUM AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

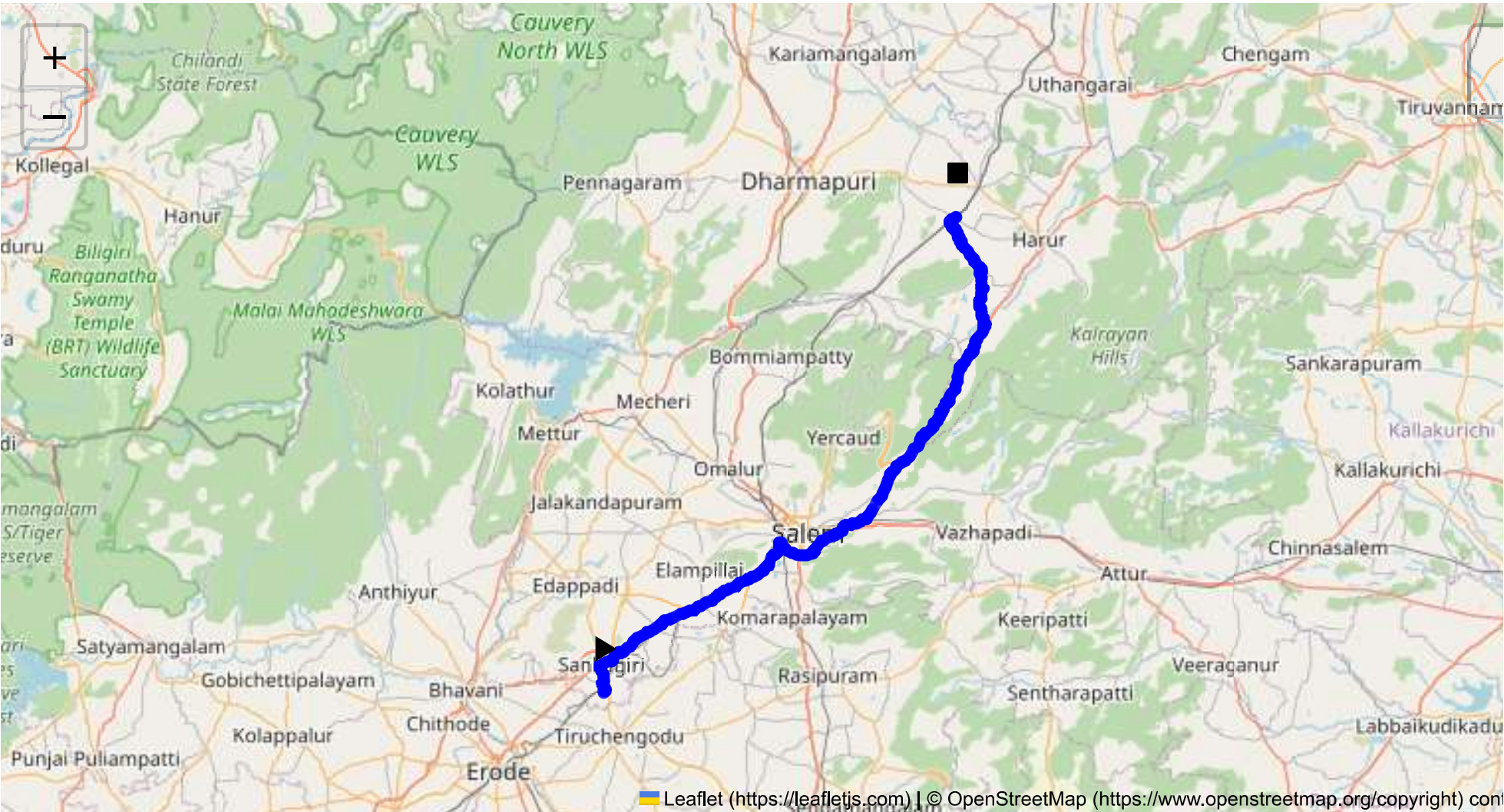
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 109.27 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (12.084899, 78.363692)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 39M7+XF Chintalpadi primarily follows regional highways and state roads. Key waypoints likely include traveling along NH44 and other connecting rural roads. This route encompasses a mix of urban, semi-urban, and rural areas.

Weather Conditions and Hazards

- **Typical Weather:** Tamil Nadu generally experiences tropical weather, with hot and humid conditions year-round. The monsoon season, from June to September, may bring heavy rains.
- **Weather-Related Hazards:** During monsoons, roads may become slippery and prone to flooding, particularly in low-lying areas.

Traffic Patterns

- **Peak Hours:** Traffic is typically heaviest in the early morning (7-9 AM) and late afternoon (4-7 PM). Urban areas near towns like Namakkal might experience congestion.
- **Congestion-Prone Areas:** Significant congestion can occur near intersections and market areas in smaller towns along the route.

Road Quality and Infrastructure

- **Road Quality:** National highways like NH44 are generally well-maintained, whereas rural roads may exhibit variations in road conditions, including potholes and narrow stretches.
- **Infrastructure:** Most highways have adequate signage, but rural stretches may lack proper lighting and road markings.

Alternative Routes

- **Suggestion for Emergencies:** An alternative might include bypassing certain rural roads by taking the broader and better-maintained stretches of NH44, where feasible, to circumvent delays due to road conditions.

Local Regulations on Hazardous Material Transport

- **Regulations:** Compliance with The Motor Vehicle Act and guidelines issued by the Petroleum and Explosives Safety Organisation (PESO) is mandatory. Restrictions may apply within city limits and near sensitive installations.

Historical Incidents

- **Historical Records:** Accidents involving heavy trucks often occur due to overloading or driver fatigue. Historical incident data point to intersections and sharp curves as common sites of such occurrences.

Environmental Considerations

- **Sensitive Areas:** Potential wildlife crossings may be present; local farmlands and water bodies near the route could be environmentally sensitive areas.

Communication Coverage

- **Coverage:** Major highways are likely to have good telecom coverage, though rural areas might experience occasional dead zones where mobile signals are weak or nonexistent.

Emergency Response Times

- **Emergency Services:** Response times could range between 30 to 60 minutes depending on proximity to major towns. Services in remote areas may take longer.

Overall Risk Assessment

- **Summary:** The route poses moderate risks primarily due to weather conditions, road quality, and regional congestion. Planning travel during non-peak hours and clear weather can mitigate some hazards. Adherence to traffic laws and safe driving practices, especially when transporting hazardous materials, will further enhance safety. Engage with local law enforcement and emergency services for route-specific information and support when necessary.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	Medium	11.45000, 77.87213	30 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.45309, 77.87325	30 KM/Hr
7	Turn	Medium	11.46400, 77.87323	30 KM/Hr
8	Turn	Medium	11.46765, 77.87037	30 KM/Hr
9	Turn	High	11.47403, 77.86841	15 KM/Hr
10	Turn	High	11.49149, 77.89601	15 KM/Hr
11	Turn	High	11.49224, 77.89601	15 KM/Hr
12	Turn	Medium	11.53501, 77.95707	30 KM/Hr
13	Turn	Medium	11.53601, 77.95856	30 KM/Hr
14	Turn	High	11.64172, 78.11817	15 KM/Hr
15	Turn	Medium	11.64224, 78.11928	30 KM/Hr
16	Turn	High	11.64195, 78.12129	15 KM/Hr
17	Turn	High	11.64120, 78.12168	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Turn	Medium	11.63948, 78.12081	30 KM/Hr
19	Turn	Medium	11.66388, 78.20672	30 KM/Hr
20	Turn	Medium	11.67129, 78.22978	30 KM/Hr
21	Turn	High	11.67120, 78.23136	15 KM/Hr
22	Turn	Medium	11.67198, 78.23148	30 KM/Hr
23	Turn	Medium	11.67440, 78.23914	30 KM/Hr
24	Turn	Medium	11.80440, 78.33325	30 KM/Hr
25	Turn	Medium	11.81235, 78.34084	30 KM/Hr
26	Turn	Medium	11.83482, 78.35294	30 KM/Hr
27	Turn	Medium	11.84738, 78.36062	30 KM/Hr
28	Turn	High	11.93905, 78.40417	15 KM/Hr
29	Turn	Medium	11.93960, 78.40395	30 KM/Hr
30	Turn	High	11.94039, 78.40200	15 KM/Hr
31	Turn	Medium	11.95570, 78.39975	30 KM/Hr
32	Turn	Medium	11.96453, 78.39648	30 KM/Hr
33	Turn	Medium	11.98782, 78.39989	30 KM/Hr
34	Turn	Medium	12.00565, 78.39738	30 KM/Hr
35	Turn	Medium	12.00984, 78.39696	30 KM/Hr
36	Turn	Medium	12.01431, 78.39858	30 KM/Hr
37	Turn	Medium	12.01550, 78.39613	30 KM/Hr
38	Turn	Medium	12.05872, 78.36659	30 KM/Hr
39	Turn	High	12.07972, 78.35570	15 KM/Hr
40	Turn	Medium	12.08022, 78.35695	30 KM/Hr
41	Turn	Medium	12.08491, 78.36247	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
21	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
22	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
23	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
24	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
25	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
26	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
27	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
28	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
30	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
32	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
34	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
20	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
31	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
33	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

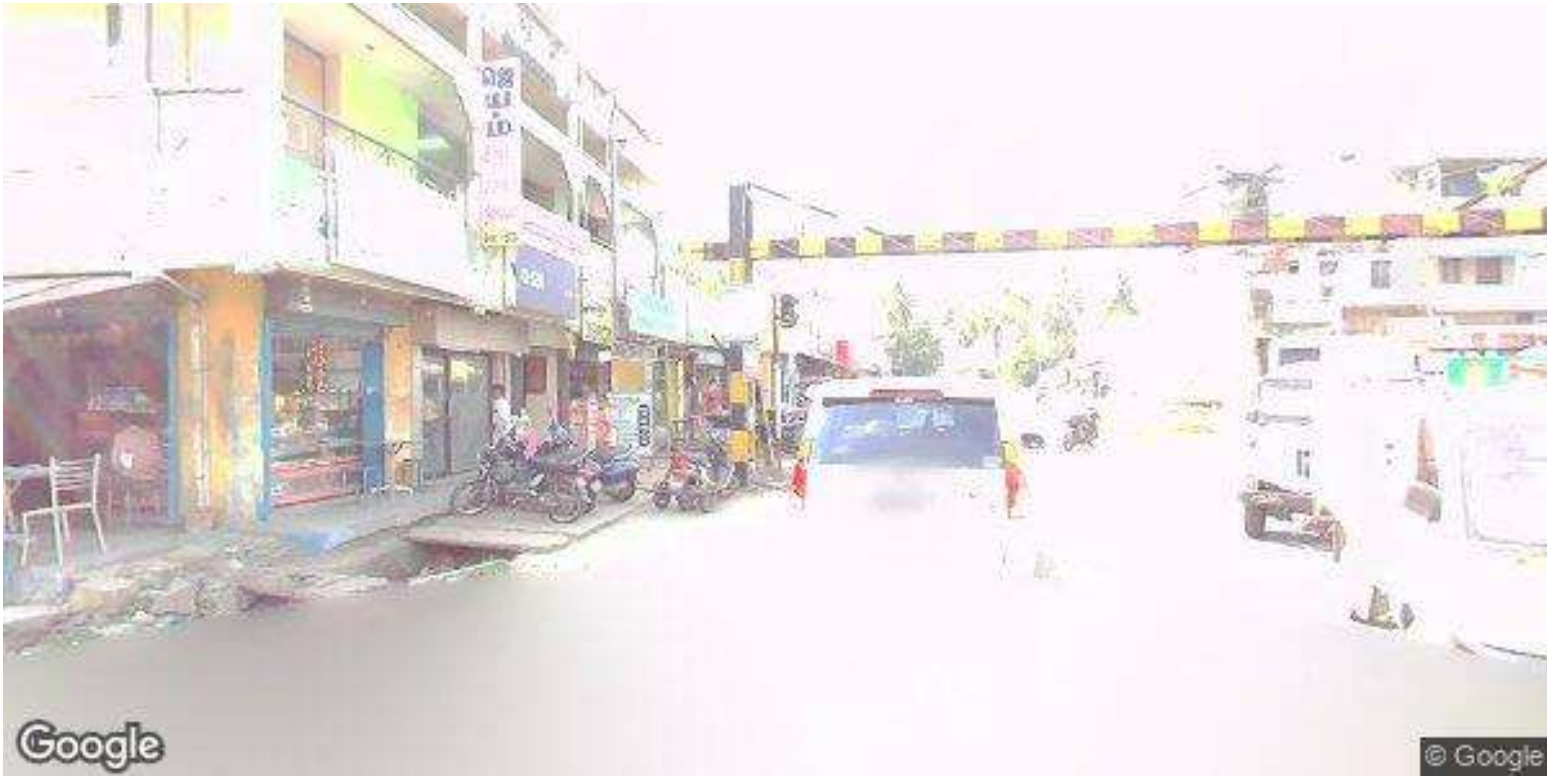


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45000, 77.87213



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46765, 77.87037



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53601, 77.95856



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64224, 78.11928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64195, 78.12129



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66388, 78.20672



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67129, 78.22978



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67198, 78.23148



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67440, 78.23914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80440, 78.33325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81235, 78.34084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83482, 78.35294



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84738, 78.36062



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93905, 78.40417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93960, 78.40395



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.94039, 78.40200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95570, 78.39975



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96453, 78.39648



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98782, 78.39989



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.00565, 78.39738



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.00984, 78.39696



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01431, 78.39858



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01550, 78.39613



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.05872, 78.36659



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.07972, 78.35570



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.08022, 78.35695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.08491, 78.36247